

HE2003 Econometrics I

Tutorial 3 (Week 4) – Problems & Solutions

Academic Year 2026/2027, Semester 1

Quantitative Research Society @NTU

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Overview

- **Lecture coverage:** Lecture 3 — the true-vs-estimated comparison (population objects vs. OLS objects), the algebraic properties of OLS residuals that follow from the first-order conditions, and the Least Squares Assumptions (with Assumption 1, zero conditional mean, as the conceptual centrepiece).
- **Questions:** 4 questions from Tutorial Question Set No. 3; solutions extend the official Tut 3 slides, which for several parts are diagram-only or cross-reference other weeks.
- **Notation:** as in lectures — u_i population error, $\hat{u}_i$ OLS residual; β_0, β_1 population (unknown) parameters, $\hat{\beta}_0, \hat{\beta}_1$ OLS estimators; “Least Squares Assumption 1/2/3” (never SLR.1–5).

Question 1 (Population vs. OLS: five core objects)

Problem

Explain the following: population regression line, OLS regression line, error term, OLS residual, and OLS predicted value. (*Tutorial Question Set No. 3, Q1.*)

Solution

Method 1: The official definitions, written out in full

Step 1: what the official slide actually says. The Tut 3 slide gives the population regression line and the error term in full, but for the last two items it simply says “*OLS residual, OLS predicted value: See Slide 17 of Week 2*” – a deliberate cross-reference to the Lecture 2 slide where those two objects were first defined, not an omission. The official text for the first three items is:

- *Population regression line.* The relationship that holds on average between X_i (e.g. average class size/STR at school district i) and Y_i (e.g. average test score at district i).
- *Error term.* u_i incorporates all the factors responsible for the difference between Y_i and the value predicted by the population regression line, $\beta_0 + \beta_1 X_i$.

Beyond the slides. Because the official answer only cross-references Slide 17 of Week 2 for the OLS residual and the OLS predicted value, we restate *all five* objects in full below, matched one-to-one against the model

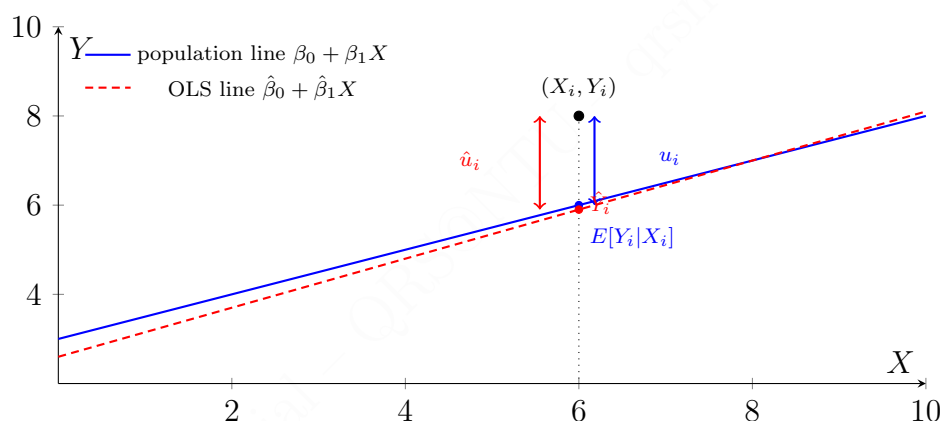
$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n.$$

1. **Population regression line:** $\beta_0 + \beta_1 X$. This is the deterministic, unknown relationship between X and Y in the population; equivalently, under Least Squares Assumption 1 (Question 3 below), it equals the conditional mean function $E[Y_i | X_i]$. Its coefficients β_0, β_1 are fixed constants of the data generating process – they do not vary from sample to sample, but they are never directly observed.
2. **Error term u_i :** $u_i = Y_i - (\beta_0 + \beta_1 X_i)$. This is the population-level deviation of Y_i from the population regression line – “all the other factors” (family background, school quality, innate ability, ...) that also affect Y_i . It is unobserved because β_0, β_1 are unknown.
3. **OLS regression line:** $\hat{\beta}_0 + \hat{\beta}_1 X$. This is the *sample-estimated* line obtained by solving $\min_{b_0, b_1} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2$ on a particular sample of n observations. It is our best guess (estimate) of the population regression line, and it is something we can actually compute.
4. **OLS predicted value (fitted value):** $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$. This is the height of the OLS regression line directly above X_i – the model’s best guess for Y_i once $\hat{\beta}_0, \hat{\beta}_1$ have been computed from the sample.
5. **OLS residual:** $\hat{u}_i = Y_i - \hat{Y}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$. This is the *observed, computable* counterpart of u_i : the vertical gap between the actual data point Y_i and the OLS-fitted value $\hat{Y}_i$. It is the sample analogue of the error term, but it is *not* numerically equal to u_i in general, because $\hat{\beta}_0, \hat{\beta}_1$ are only estimates of β_0, β_1 .

Comparison table. Every population object below has an exact sample counterpart, obtained by replacing the unknown (unhatted) quantity with its OLS-estimated (hatted) version.

Population object (unknown, fixed)	Sample counterpart (computable, varies with the sample)
Parameters β_0, β_1	OLS estimators $\hat{\beta}_0, \hat{\beta}_1$
Population regression line $\beta_0 + \beta_1 X$	OLS regression line $\hat{\beta}_0 + \hat{\beta}_1 X$
Error term $u_i = Y_i - (\beta_0 + \beta_1 X_i)$	OLS residual $\hat{u}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$
Conditional mean $E[Y_i X_i] = \beta_0 + \beta_1 X_i$ (holds under Assumption 1)	OLS predicted value $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$

Figure: population line, OLS line, u_i , and $\hat{u}_i$ at the same point. The figure below plots both lines for the same sample and marks a single observation (X_i, Y_i) : the vertical gap from Y_i down to the population line is u_i (population, unobserved), while the vertical gap from Y_i down to the OLS line is $\hat{u}_i$ (sample, computable). The two gaps differ because the two lines differ.



Interpretation. The left column of the table (population line, u_i , $E[Y_i | X_i]$, β_0, β_1) consists of theoretical objects that exist regardless of whether data has ever been collected – they describe the true, unknown data-generating process. The right column (OLS line, $\hat{u}_i$, $\hat{Y}_i$, $\hat{\beta}_0, \hat{\beta}_1$) consists of objects we can actually *compute* once we have a sample: they are our best estimates of their population counterparts, and – because they are computed from one particular sample of n observations – they will generally take different numerical values if we draw a different sample (sampling variation). The population objects, by contrast, are fixed constants of the model and do not vary with the sample.

Population: $\beta_0 + \beta_1 X_i$, $u_i = Y_i - (\beta_0 + \beta_1 X_i)$ (unobserved)
Sample: $\hat{\beta}_0 + \hat{\beta}_1 X_i$, $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$, $\hat{u}_i = Y_i - \hat{Y}_i$ (all computable)

Question 2 (Residual properties from the FOCs)

Problem

- (a) Show that $\sum_{i=1}^n \hat{u}_i = 0$ by using the first-order condition (FOC) of the OLS estimators.
- (b) Show that $\sum_{i=1}^n \hat{u}_i X_i = 0$ by using the FOC of the OLS estimators.
- (c) Show that $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$. [Hint: $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \sum_{i=1}^n \hat{u}_i (\hat{\beta}_0 + \hat{\beta}_1 X_i)$.]

(Tutorial Question Set No. 3, Q2.)

Solution

Method 1: First-order conditions (official)

Step 1: recall the minimization problem. The OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ solve

$$\min_{b_0, b_1} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2.$$

Taking partial derivatives with respect to b_0 and b_1 and evaluating at $(\hat{\beta}_0, \hat{\beta}_1)$ gives the two FOCs:

$$\frac{\partial}{\partial b_0} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2 \Big|_{(\hat{\beta}_0, \hat{\beta}_1)} = -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad (1)$$

$$\frac{\partial}{\partial b_1} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2 \Big|_{(\hat{\beta}_0, \hat{\beta}_1)} = -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0. \quad (2)$$

Step 2: part (a). Since $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ by definition, dividing (1) by -2 gives directly

$$\boxed{\sum_{i=1}^n \hat{u}_i = 0.}$$

Step 3: part (b). Dividing (2) by -2 gives directly

$$\boxed{\sum_{i=1}^n \hat{u}_i X_i = 0.}$$

Step 4: part (c), using the hint. Using $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$,

$$\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \sum_{i=1}^n \hat{u}_i (\hat{\beta}_0 + \hat{\beta}_1 X_i) = \hat{\beta}_0 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^n \hat{u}_i X_i.$$

By parts (a) and (b), both sums on the right are zero, so

$$\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \hat{\beta}_0(0) + \hat{\beta}_1(0) = 0.$$

Method 2: Direct substitution of the closed-form estimators (parts (a), (b))

This route never invokes the minimization problem at all; it uses only the closed-form formulas

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}, \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

derived in Tutorial 2, together with the summation identity $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}$ (also from Tutorial 2, Q4).

Part (a) by substitution.

$$\begin{aligned} \sum_{i=1}^n \hat{u}_i &= \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = \sum_{i=1}^n Y_i - n\hat{\beta}_0 - \hat{\beta}_1 \sum_{i=1}^n X_i \\ &= n\bar{Y} - n(\bar{Y} - \hat{\beta}_1 \bar{X}) - \hat{\beta}_1 (n\bar{X}) \\ &= n\bar{Y} - n\bar{Y} + n\hat{\beta}_1 \bar{X} - n\hat{\beta}_1 \bar{X} = 0. \end{aligned}$$

No FOC is used here – only the definition of $\hat{\beta}_0$ in terms of $\bar{Y}$, $\bar{X}$, $\hat{\beta}_1$.

Part (b) by substitution.

$$\begin{aligned} \sum_{i=1}^n \hat{u}_i X_i &= \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = \sum_{i=1}^n X_i Y_i - \hat{\beta}_0 \sum_{i=1}^n X_i - \hat{\beta}_1 \sum_{i=1}^n X_i^2 \\ &= \sum_{i=1}^n X_i Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X})(n\bar{X}) - \hat{\beta}_1 \sum_{i=1}^n X_i^2 \\ &= \left[\sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y} \right] - \hat{\beta}_1 \left[\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right] \\ &= \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) - \hat{\beta}_1 \sum_{i=1}^n (X_i - \bar{X})^2. \end{aligned}$$

But $\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$, so $\hat{\beta}_1 \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$ exactly. Substituting,

$$\sum_{i=1}^n \hat{u}_i X_i = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) - \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = 0.$$

Interpretation. In-sample, the OLS residuals have sample mean zero ($\frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0$) and zero sample covariance with X_i (since $\frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0$, the term $\frac{1}{n} \sum_{i=1}^n \hat{u}_i X_i = 0$ is exactly that covariance). Part (c) extends this: the residuals also have zero sample covariance with the fitted values $\hat{Y}_i$. All three are properties of the fitted line itself, not of the population model.

Beyond the slides – a common confusion. These three identities are *algebraic facts about any least-squares fit*: they follow purely from the FOCs (or, equivalently, from the closed-form formulas for $\hat{\beta}_0, \hat{\beta}_1$), and they hold for *every* sample, whether or not Least Squares Assumption 1 ($E[u_i | X_i] = 0$, Question 3) is true of the population. Do not confuse the sample identity $\sum_{i=1}^n \hat{u}_i = 0$ (always true, mechanical) with the population statement $E[u_i] = 0$ or $E[u_i | X_i] = 0$ (an assumption about the data-generating process that may fail). OLS forces the residuals to satisfy $\sum_{i=1}^n \hat{u}_i = 0$ and $\sum_{i=1}^n \hat{u}_i X_i = 0$ by construction, in exactly the same way that OLS forces the fitted line through $(\bar{X}, \bar{Y})$ – it says nothing about whether the population error u_i genuinely has mean zero given X_i .

Technique note. Any question asking you to prove a property of $\hat{u}_i$ that involves only $\hat{u}_i$, X_i , or constants can usually be attacked two ways: (i) read it straight off the FOCs (fastest when the FOC has exactly the required shape), or (ii) substitute the closed-form $\hat{\beta}_0, \hat{\beta}_1$ and simplify using the Tutorial 2 summation identities (useful when the FOC is not literally in the form you need, or as an independent check).

$$\sum_{i=1}^n \hat{u}_i = 0, \quad \sum_{i=1}^n \hat{u}_i X_i = 0, \quad \sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0.$$

Question 3 (Assumption 1 by diagrams, and a counterexample)

Problem

- (a) Explain the meaning of the OLS Assumption 1 by using diagrams.
- (b) Explain how we can obtain the figure shown on slide No. 11 for Lecture 3.
- (c) Give an example where $E[u] = 0$ but $E[u | X] \neq 0$ by using diagrams.

(Tutorial Question Set No. 3, Q3.)

Solution

Method 1: Reading the assumption directly off the figures (official route)

(a) **The meaning of Least Squares Assumption 1.** Assumption 1 states

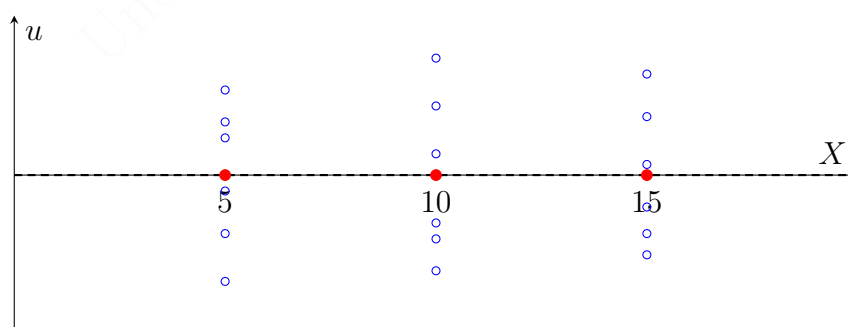
$$E[u_i | X_i] = 0,$$

i.e. at *every* value of X , the population-average value of the error term is zero. The official figure (reproduced below as Figure 1) plots u_i against X_i and highlights three fixed values $X = 5, 10, 15$: at each of these values, some sampled error terms lie above zero ($u_i > 0$) and some lie below zero ($u_i < 0$), but their average at that particular X is exactly zero. Formally,

$$E[u_i | X_i = 5] = 0, \quad E[u_i | X_i = 10] = 0, \quad E[u_i | X_i = 15] = 0, \quad \dots,$$

and this holds simultaneously for every value X can take – compactly written $E[u_i | X_i] = 0$.

Figure 1: u_i vs. X_i – zero conditional mean at $X = 5, 10, 15$



(b) **Obtaining the Lecture 3 slide-11 figure.** Slide 11's figure plots Y_i against X_i , with the population regression line drawn in and the data shown scattered around it at $X = 5, 10, 15, \dots$; we now derive *why* that is the correct picture, starting from Figure 1.

Step 1. Start from the regression equation, $Y_i = \beta_0 + \beta_1 X_i + u_i$.

Step 2. Condition on X_i and take expectations of both sides. Since β_0, β_1 are constants and X_i is being conditioned on, linearity of conditional expectation gives

$$E[Y_i | X_i] = E[\beta_0 | X_i] + E[\beta_1 X_i | X_i] + E[u_i | X_i] = \beta_0 + \beta_1 X_i + E[u_i | X_i].$$

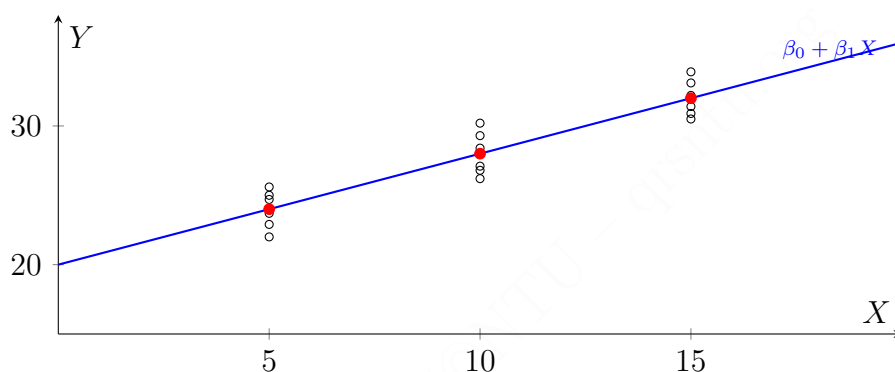
Step 3. Substitute Assumption 1, $E[u_i | X_i] = 0$:

$$E[Y_i | X_i] = \beta_0 + \beta_1 X_i.$$

So, given any value of X , the population-average value of Y lies *exactly* on the population regression line – we do not need to know the realised value of u_i , because it cancels out on average.

Step 4. Geometrically, this means the slide-11 figure (Figure 2 below) is obtained from Figure 1 by *shifting every point up by the height of the population line at its own X value*: a point that was u_i above the zero line in Figure 1 becomes a point that is u_i above $\beta_0 + \beta_1 X_i$ in Figure 2, i.e. it becomes $Y_i = \beta_0 + \beta_1 X_i + u_i$. Because the u_i 's still average to zero at each X (Assumption 1), the Y_i 's at each X still average to a point exactly on the population line.

Figure 2: Y_i vs. X_i – data centred on the population line at $X = 5, 10, 15$



The red dots in Figure 2 – the conditional means $E[Y_i | X_i = 5, 10, 15]$ – sit exactly on the blue population line, by the derivation above; this is precisely the content of slide 11.

(c) A counterexample: $E[u] = 0$ but $E[u | X] \neq 0$ (**official**). For simplicity, consider X taking only two values, $X \in \{0, 1\}$ (as in Question 4 below), with $\Pr(X = 1) = \Pr(X = 0) = \frac{1}{2}$. Suppose

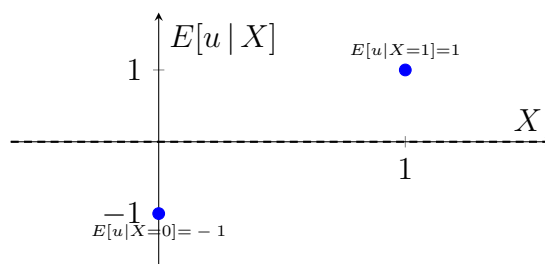
$$E[u | X = 0] = -1 < 0, \quad E[u | X = 1] = 1 > 0.$$

By the Law of Total Expectation,

$$E[u] = \Pr(X = 0) E[u | X = 0] + \Pr(X = 1) E[u | X = 1] = \frac{1}{2}(-1) + \frac{1}{2}(1) = 0,$$

so the *unconditional* mean of u is exactly zero. Yet $E[u | X]$ is *not* identically zero – it equals -1 at $X = 0$ and $+1$ at $X = 1$ – so Assumption 1 fails even though $E[u] = 0$.

Figure 3: binary X – $E[u] = 0$ but $E[u | X] \neq 0$



Method 2 (Beyond the slides): a continuous counterexample

The official slide only gives the binary- X example above. Here is a genuinely different example with a *continuous* regressor, which shows the same phenomenon does not depend on X being discrete.

Setup. Let $X \sim N(0, 1)$ and define $u = X^2 - 1$.

Check $E[u] = 0$. Since $\text{Var}(X) = E[X^2] - (E[X])^2$ and $X \sim N(0, 1)$ has $E[X] = 0$, $\text{Var}(X) = 1$,

$$E[X^2] = \text{Var}(X) + (E[X])^2 = 1 + 0 = 1 \implies E[u] = E[X^2] - 1 = 1 - 1 = 0.$$

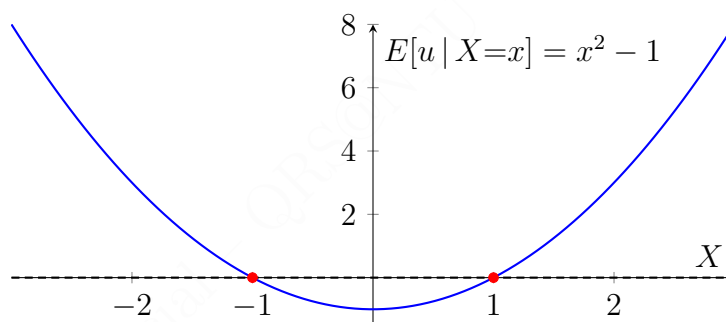
So the unconditional mean of u is exactly zero.

Check $E[u | X] \neq 0$. Conditioning on $X = x$ makes x a known constant, so

$$E[u | X = x] = E[X^2 - 1 | X = x] = x^2 - 1,$$

which is *not* identically zero – it is zero only at the two isolated points $x = \pm 1$, and strictly positive (a parabola) everywhere else. Hence $E[u | X] = X^2 - 1 \neq 0$ as a function of X , even though $E[u] = 0$.

Figure 4: continuous counterexample, $u = X^2 - 1$, $X \sim N(0, 1)$



Interpretation. Both counterexamples make the same point: $E[u | X] = 0$ is a *much stronger* requirement than $E[u] = 0$. The unconditional mean is a single number averaged over the whole population of X -values; it can vanish purely because positive and negative deviations at different values of X cancel out *across* X . Assumption 1 requires the average to vanish *separately, at every single value of* X – no cancellation across X is allowed. This is precisely the distinction the question is designed to test, and it is a very common source of confusion: students sometimes verify only that $E[u] = 0$ (which OLS guarantees automatically for the *residuals*, by Question 2(a)) and mistakenly conclude that Assumption 1 must hold.

Assumption 1: $E[u_i | X_i] = 0$ for every value of $X_i \implies E[Y_i | X_i] = \beta_0 + \beta_1 X_i$.

Counterexamples: binary X with $E[u|X=0]=-1$, $E[u|X=1]=1$;

continuous $u = X^2 - 1$, $X \sim N(0, 1)$ – both have $E[u] = 0$ but $E[u|X] \neq 0$.

Question 4 (Binary regressor: SSS vs. SOH wages)

Problem

Suppose we want to find out if there exists a difference in average wages between graduates from the School of Social Sciences (SSS) and graduates from the School of Humanities (SOH), using the linear regression model with an indicator variable ($X = 1$ or 0).

- Write down the linear regression model for this problem. Define the dependent variable, explanatory variable, and error term in your model.
- Write down the corresponding OLS Assumption 1. Explain in what case the OLS Assumption 1 is likely to hold/unlikely to hold.
- Now suppose that the OLS Assumption 1 does hold; explain the specific meaning of the parameters β_0 and β_1 using conditional expectation. [Hint: derive the conditional expectation $E[Y_i | X_i]$ for $X_i = 1$ and $X_i = 0$, respectively.]
- Illustrate the meaning of β_0 and β_1 by using a diagram.

(Tutorial Question Set No. 3, Q4.)

Solution

Method 1: Conditional-expectation route (official)

(a) The model.

$$\text{wage}_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n,$$

where:

- wage_i is the **dependent variable**: the wage of graduate i ;
- X_i is the **explanatory variable (regressor)**, a binary indicator with $X_i = 1$ if individual i graduated from SSS and $X_i = 0$ if individual i graduated from SOH;
- u_i is the **error term**, capturing all other factors that may affect individual i 's wage (e.g. innate ability, family background, work experience, effort).

(b) Assumption 1 for this model. Least Squares Assumption 1, $E[u_i | X_i] = 0$, is here equivalent (since X_i takes only the two values $0, 1$) to requiring the conditional mean to vanish at *each* value separately:

$$E[u_i | X_i] = 0 \iff E[u_i | X_i = 1] = E[u_i | X_i = 0] (= 0).$$

In words: the average of the "other factors" (loosely, ability) among SSS graduates equals the average among SOH graduates – neither group is systematically more or less able (or otherwise advantaged) than the other, once we control for the school indicator.

- Likely to hold*: if students were *randomly assigned* to SSS or SOH (as in an RCT), assignment would be independent of ability and all other wage-relevant traits, so the two groups would be balanced on average and Assumption 1 would hold.
- Likely to fail*: in practice, admission is not random. Students *self-select* into SSS or SOH based on ability, interests, or academic preparation, and the two groups may come from systematically different family backgrounds or secondary-school tracks. Any such

sorting that is correlated with wage-relevant traits (and with the school choice X_i) breaks Assumption 1.

(c) Meaning of β_0 and β_1 . Assume Assumption 1 holds. Exactly as derived in Question 3(b),

$$E[Y_i | X_i] = E[\beta_0 + \beta_1 X_i + u_i | X_i] = \beta_0 + \beta_1 X_i + E[u_i | X_i] = \beta_0 + \beta_1 X_i.$$

Evaluating at the two possible values of X_i :

$$E[Y_i | X_i = 1] = \beta_0 + \beta_1, \quad E[Y_i | X_i = 0] = \beta_0.$$

Subtracting,

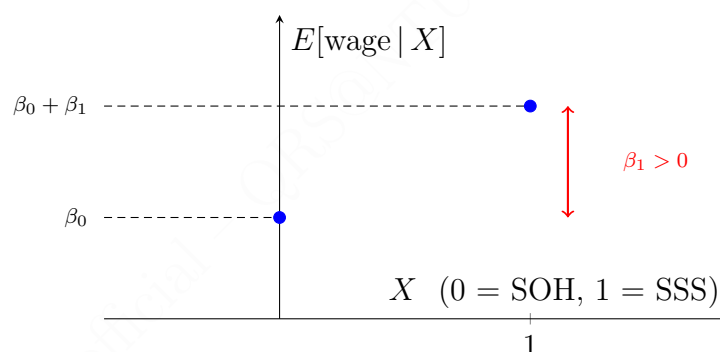
$$\beta_1 = E[Y_i | X_i = 1] - E[Y_i | X_i = 0].$$

Hence:

- β_0 is the population **average wage of SOH graduates** ($X = 0$);
- β_1 is the **difference in average wages** between SSS graduates and SOH graduates.

Because the regressor is binary, β_1 is *not* a slope or a rate of change (there is no continuum of X values along which Y moves smoothly) – it is a pure **level shift** between the two group means.

(d) Diagram. The diagram below marks the two conditional means, β_0 at $X = 0$ and $\beta_0 + \beta_1$ at $X = 1$, for the case $\beta_1 > 0$ (SSS graduates earn more on average).



Method 2 (Beyond the slides): the OLS sample estimates

The official slides stop at the population-level interpretation of β_0, β_1 . We now show the exact sample analogue: with a binary regressor, the OLS estimators reduce to simple sample group means, mirroring the population result exactly.

Setup. Let n_1 be the number of sample observations with $X_i = 1$ (SSS) and $n_0 = n - n_1$ the number with $X_i = 0$ (SOH), so $\bar{X} = n_1/n$. Let $\bar{Y}_1$ denote the sample mean of wage among the SSS observations and $\bar{Y}_0$ the sample mean among the SOH observations, so that $n\bar{Y} = n_1\bar{Y}_1 + n_0\bar{Y}_0$.

Derivation of $\hat{\beta}_1$. Since $X_i \in \{0, 1\}$, we have $\sum_{i=1}^n X_i Y_i = \sum_{X_i=1} Y_i = n_1 \bar{Y}_1$. (Note that the $X_i = 0$ observations do *not* drop out of $\sum_{i=1}^n (X_i - \bar{X}) Y_i$: each carries the weight $-\bar{X} \neq 0$.) Splitting the sum accordingly:

$$\sum_{i=1}^n (X_i - \bar{X}) Y_i = \sum_{i=1}^n X_i Y_i - \bar{X} \sum_{i=1}^n Y_i = n_1 \bar{Y}_1 - \bar{X} (n \bar{Y}) = n_1 \bar{Y}_1 - n_1 \bar{Y} = n_1 (\bar{Y}_1 - \bar{Y}).$$

Also $\sum_{i=1}^n (X_i - \bar{X})\bar{Y} = \bar{Y} \sum_{i=1}^n (X_i - \bar{X}) = 0$, so $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n (X_i - \bar{X})Y_i = n_1(\bar{Y}_1 - \bar{Y})$. Using $\bar{Y} = (n_1\bar{Y}_1 + n_0\bar{Y}_0)/n$,

$$\begin{aligned}\bar{Y}_1 - \bar{Y} &= \frac{n\bar{Y}_1 - n_1\bar{Y}_1 - n_0\bar{Y}_0}{n} = \frac{n_0(\bar{Y}_1 - \bar{Y}_0)}{n} \\ \Rightarrow \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) &= \frac{n_1n_0(\bar{Y}_1 - \bar{Y}_0)}{n}.\end{aligned}$$

For the denominator, $\sum_{i=1}^n (X_i - \bar{X})^2 = n_0\bar{X}^2 + n_1(1 - \bar{X})^2$, and substituting $\bar{X} = n_1/n$ simplifies this to n_1n_0/n . Therefore

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{n_1n_0(\bar{Y}_1 - \bar{Y}_0)/n}{n_1n_0/n} = \bar{Y}_1 - \bar{Y}_0.$$

Derivation of $\hat{\beta}_0$.

$$\begin{aligned}\hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1\bar{X} = \frac{n_1\bar{Y}_1 + n_0\bar{Y}_0}{n} - (\bar{Y}_1 - \bar{Y}_0)\frac{n_1}{n} \\ &= \frac{n_1\bar{Y}_1 + n_0\bar{Y}_0 - n_1\bar{Y}_1 + n_1\bar{Y}_0}{n} = \frac{(n_0 + n_1)\bar{Y}_0}{n} = \bar{Y}_0.\end{aligned}$$

Result.

$$\hat{\beta}_0 = \bar{Y}_0, \quad \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0.$$

This is the *exact sample mirror* of the population result in part (c): $\hat{\beta}_0$ is simply the sample mean SOH wage, and $\hat{\beta}_1$ is simply the difference between the sample mean SSS wage and the sample mean SOH wage. No calculus beyond the ordinary OLS formula is required – the binary regressor collapses the general formula into a comparison of two group averages.

Interpretation. This is exactly the population-vs-sample duality of Question 1: β_0, β_1 (population, defined via $E[Y | X]$) are unknown constants; $\hat{\beta}_0, \hat{\beta}_1$ (sample, defined via the OLS formula) are the computable group-mean statistics $\bar{Y}_0$ and $\bar{Y}_1 - \bar{Y}_0$ that estimate them. If Assumption 1 holds (random assignment to schools), $\hat{\beta}_0, \hat{\beta}_1$ are good (unbiased) estimates of the true wage level and wage gap; if it fails (self-selection by ability), the estimated gap $\hat{\beta}_1$ will confound the true causal wage effect of attending SSS with any pre-existing ability difference between the two groups.

$\text{wage}_i = \beta_0 + \beta_1 X_i + u_i, \quad X_i = 1 \text{ (SSS), } 0 \text{ (SOH)}$ Assumption 1: $E[u_i X_i=1] = E[u_i X_i=0] = 0$; holds under random assignment, fails under self-selection by ability/background $\beta_0 = E[\text{wage} \text{SOH}], \quad \beta_1 = E[\text{wage} \text{SSS}] - E[\text{wage} \text{SOH}] \text{ (not a slope)}$ Beyond the slides: $\hat{\beta}_0 = \bar{Y}_0, \quad \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$
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Method Map

Q.	Core topic	Methods included
1	Population vs. OLS objects	Restating the five official definitions in full; comparison table; pgfplots figure of both lines with u_i and $\hat{u}_i$ marked at the same point.
2	Residual properties from OLS	FOC reading (official); direct substitution of the closed-form $\hat{\beta}_0, \hat{\beta}_1$ (parts (a),(b) only); hint-based combination for part (c).
3	Assumption 1 via diagrams	Reading zero conditional mean off the u -vs- X figure; deriving $E[Y X] = \beta_0 + \beta_1 X$ to obtain the slide-11 figure; binary- X counterexample (official) and a continuous counterexample $u = X^2 - 1$, $X \sim N(0, 1)$ (beyond the slides).
4	Binary regressor (SSS vs. SOH)	Conditional-expectation route for β_0, β_1 (official); diagram; OLS sample estimates $\hat{\beta}_0 = \bar{Y}_0$, $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$ derived from the closed-form formula (beyond the slides).

Technique Summary

- Population vs. sample duality.** Every population object has an exact sample counterpart obtained by “hatting” it: $\beta \rightarrow \hat{\beta}$, $u_i \rightarrow \hat{u}_i$, $\beta_0 + \beta_1 X \rightarrow \hat{\beta}_0 + \hat{\beta}_1 X$, $E[Y_i|X_i] \rightarrow \hat{Y}_i$. Question 1’s comparison table is the master key for keeping these six objects straight.
- FOC identities are sample facts, not population assumptions.** $\sum_{i=1}^n \hat{u}_i = 0$, $\sum_{i=1}^n \hat{u}_i X_i = 0$, and $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$ hold algebraically for *any* least-squares fit, regardless of whether Assumption 1 ($E[u_i|X_i] = 0$) is true of the population. Never cite the residual identities as evidence that Assumption 1 holds.
- Zero conditional mean $\neq$ zero unconditional mean.** $E[u|X] = 0$ must hold at *every* value of X separately; $E[u] = 0$ is a single population-wide average that can vanish purely through cancellation across different values of X . Question 3(c) supplies two worked counterexamples (binary and continuous X) where the latter holds but not the former.
- One derivation, two figures.** $E[Y_i|X_i] = \beta_0 + \beta_1 X_i$ follows in two lines from linearity of conditional expectation plus Assumption 1; this single derivation both explains how the “data centred on the line” figure is obtained from the “errors centred on zero” figure, and pins down the interpretation of β_0, β_1 with a binary regressor.
- Binary regressors: means, not slopes.** With $X_i \in \{0, 1\}$, β_1 is the *difference in conditional means* between the $X = 1$ and $X = 0$ groups, never a rate of change; this carries over exactly to the OLS sample estimates, which collapse to $\hat{\beta}_0 = \bar{Y}_0$ and $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$.

Final Answers

1. Population regression line $\beta_0 + \beta_1 X$ (unobserved, fixed); error term $u_i = Y_i - (\beta_0 + \beta_1 X_i)$ (unobserved); OLS regression line $\hat{\beta}_0 + \hat{\beta}_1 X$ (computed from a sample); OLS predicted value $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$; OLS residual $\hat{u}_i = Y_i - \hat{Y}_i$ (computed). See the comparison table and figure in Question 1.
2. $\sum_{i=1}^n \hat{u}_i = 0$; $\sum_{i=1}^n \hat{u}_i X_i = 0$; $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$ (all three follow from the FOCs and hold for any OLS fit).
3. (a) Assumption 1 means $E[u_i|X_i] = 0$ at every value of X (Figure 1). (b) The slide-11 figure follows from $E[Y_i|X_i] = \beta_0 + \beta_1 X_i$, derived from Assumption 1 and linearity of conditional expectation (Figure 2). (c) Counterexamples: binary X with $E[u|X=0] = -1$, $E[u|X=1] = 1$ (Figure 3); continuous $u = X^2 - 1$ with $X \sim N(0, 1)$, giving $E[u] = 0$ but $E[u|X] = X^2 - 1 \neq 0$ (Figure 4).
4. (a) $\text{wage}_i = \beta_0 + \beta_1 X_i + u_i$, $X_i = 1$ (SSS) or 0 (SOH). (b) Assumption 1: $E[u_i|X_i=1] = E[u_i|X_i=0] = 0$; plausible under random assignment to schools, implausible under self-selection by ability or family background. (c) $\beta_0 = E[\text{wage}|\text{SOH}]$, $\beta_1 = E[\text{wage}|\text{SSS}] - E[\text{wage}|\text{SOH}]$ (a mean difference, not a slope). (d) Diagram with β_0 at $X = 0$ and $\beta_0 + \beta_1$ at $X = 1$. Beyond the slides: $\hat{\beta}_0 = \bar{Y}_0$, $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$.